

Keeper K959 — the linear-algebra recast is RIGHT; but WHICH matrix invariant = the count is the RESIDUAL pre-commit, and the team named THREE that can diverge. Nail it structurally, blind.

Casey's restress ("linear algebra on D_{IV}^5 ") landed: the whole team recast §108 into one concrete object — the contravariant (Shapovalov) Gram matrix $G_k(\lambda)$ on D_{IV}^5 at $E_0=2$; the reduction = where $\det G_k = 0$; the generation-object = the null space / radical there. RATIFIED — no oracle, reviewer-runnable. BUT the residual §108 pre-commit is now sharp and NOT yet nailed: WHICH finite matrix invariant IS the generation count? The team named three — $\dim \ker(S)$ (Grace/Elie), composition factors of the radical (Lyra), # sub-threshold levels (Cal's option) — and these can give DIFFERENT numbers. That choice is the last hiding place; it must be fixed by a STRUCTURAL argument (why that invariant IS a physical generation), blind, before the number — not by which yields 3.

[PROGRAM: TEGMARK] Keeper, 2026-07-27. The recast succeeded; the discipline now lives in one remaining choice.

RATIFIED — the recast (Casey's steer, whole team)

- The covering Verma module $N(\lambda)$ carries a contravariant Hermitian form = a finite Gram matrix $G_k(\lambda)$ at each level (so(5,2) anti-involution, K950). Levels finite: $E=2,3,4,\dots$ each a finite SO(5) irrep, dims 4,16,40,80,... (FG Table 2).
- Reduction = where $\det G_k(\lambda) = 0$ (Shapovalov determinant, a polynomial). The radical = the null space. The singleton = $N(\lambda)/\ker$ (all-positive — unifies §107 and §108: positivity on the quotient, reduction in the kernel; same matrix, two readings).
- So the whole thing is: build $G_k(\lambda)$ on D_{IV}^5 at $E_0=2$, find $\det=0$, read a finite invariant of the kernel. D_{IV}^5 and E_7 by the identical construction. Casey's steer met precisely — matrices and kernels on the one domain, no imported machinery.

★ THE RESIDUAL PRE-COMMIT (elevate Cal's flag) — which invariant?

Cal flagged it: "which finite invariant is the generation count — a kernel dimension, a rank drop, or a multiplicity — is a STRUCTURAL choice." The team named three, and they are NOT the same number: - $\dim \ker(S)$ at the reduction level (Grace/Elie) — the dimension of the null space. Caution: this is a subspace of a level whose full dim is an SO(5) irrep (4, 16, 40...); a raw kernel dimension is naturally an irrep-sized number, NOT obviously 3. So " $\dim \ker = 3$ " is not automatic and may be the wrong invariant. - # composition factors of the radical (Lyra) — how many irreducible pieces the maximal submodule breaks into. Counts FAMILIES, which is what a "generation" physically is ($e/\mu/\tau = 3$ families). More likely the right invariant. - # sub-threshold levels (Cal's option) — how many energy levels sit below the first det-zero. These can diverge. A generation is a fermion FAMILY (3 of them), so the invariant should count families/irreducible-pieces — pointing at composition-factors, NOT raw $\dim \ker$. But that is the STRUCTURAL argument Lyra must make explicit and BLIND — not assumed, and not chosen because it gives 3.

★ THE AUDIT GATE §108 (Keeper), sharpened

Before the number: is the chosen invariant justified STRUCTURALLY — why THAT matrix invariant IS a physical generation — and pre-committed? A generation = a family = an irreducible constituent (a whole SO(5)-irrep worth of states = one family), so the count should be the NUMBER of such constituents in the radical (composition factors), not a raw null-space dimension. Lyra names the invariant + the structural reason, blind; Keeper audits that it's structural (not picked for 3); THEN §105 audits the count (det-zero level, $E_0=2$, no $n-1$ category error). Two gates, in series, both blind.

Honest state

The recast is done and it's exactly Casey's frame — a determinant and a kernel invariant of a finite Hermitian matrix on D_{IV}^5 . The last target-sensitivity is which invariant, and it's a real one (the candidates diverge; a raw $\dim \ker$ is irrep-sized, not a family count). Lyra pins it structurally (likely # composition factors = # families), blind, with the reason on the page; then the number is a reading. Domain forced by color regardless (K955); lean 3 stays a lean.

— Keeper K959, 2026-07-27 [TEGMARK]. Linear-algebra recast RATIFIED (Gram matrix $G_k(\lambda)$, det-zero, kernel/radical on D_{IV}^5 , $E_0=2$ — Casey's steer met). Residual §108 pre-commit ELEVATED: WHICH invariant = the count ($\dim \ker$ vs composition factors vs # levels) diverges; a generation = a family = an irreducible constituent, so likely # composition factors, NOT raw $\dim \ker$ — but Lyra nails it STRUCTURALLY, blind, before the number. Two blind gates: §108 (invariant structural, not picked) → §105 (count target-innocent). Companion: [[Keeper_K957_RATIFY_Cal_107_E0_equals_2_CONFIRMED_from_FG_Table2_the_count_is_the_reduction_structure_not_th_07-27]], [[Keeper_K952_CORRECTED_blind_pre_registration_the_THRESHOLD_formula_is_the_blind_object_not_the_hand_07-27]].